No.	Status	Date
LNDE-RPT-96-014-DS-4-RA-101	Issued	1996-03-08

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 4-RA-101, which carries rich amine within the ATU-100. The scope is limited to the baseline thickness recording of the specified line to establish initial wall thickness data prior to the unit entering service.

Grace Okafor, acting as the UT technician, is responsible for the execution of the measurements and the recording of the data. All activities were performed in accordance with the requirements of the SNT-TC-1A UT Level II certification.

Technical Basis

The thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 0. The methodology for the ultrasonic testing was conducted in accordance with the standard ASTM E797.

The data recorded herein forms part of the larger baseline survey detailed in LNDE-RPT-96-014, which covers 118 CMLs for the ATU-100.

Line Specifications

Line 4-RA-101 is constructed from A106 Gr.B with an outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design pressure is 862 kPa(g) and the design temperature is 82 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. This value serves as the retirement thickness for the asset.

Measurement Data

Thickness readings were taken at the designated locations for 4-RA-101. As the unit is not yet in service, these readings establish the baseline for future corrosion rate calculations.

All measurements are compared against the t-min of 3.81 mm to ensure the line meets the required thickness specifications prior to startup.

Thickness readings, measured 1996-03-08

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	—	—	7.12	—	—	—	2001-03-08	
6-SG-101-02	7.11	3.56	—	—	7.05	—	—	—	2001-03-08	
6-SG-101-03	7.11	3.56	—	—	7.08	—	—	—	2001-03-08	
6-SG-101-04	7.11	3.56	—	—	7.09	—	—	—	2001-03-08	
6-SG-101-05	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-101-06	7.11	3.56	—	—	7.07	—	—	—	2001-03-08	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-102-02	7.11	3.56	—	—	7.11	—	—	—	2001-03-08	
6-SG-102-03	7.11	3.56	—	—	7.04	—	—	—	2001-03-08	
6-SG-102-04	7.11	3.56	—	—	6.99	—	—	—	2001-03-08	
6-SG-102-05	7.11	3.56	—	—	7.13	—	—	—	2001-03-08	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-02	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-03	8.18	4.06	—	—	8.23	—	—	—	2006-03-08	
8-TG-101-04	8.18	4.06	—	—	8.10	—	—	—	2006-03-08	
8-TG-101-05	8.18	4.06	—	—	8.22	—	—	—	2006-03-08	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-101-02	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-101-04	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-101-05	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-06	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-101-07	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	—	—	6.02	—	—	—	2006-03-08	
4-LA-102-02	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-03	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-04	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-102-05	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-06	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-07	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	—	—	7.02	—	—	—	2006-03-08	
6-LA-103-02	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-03	7.11	3.56	—	—	7.08	—	—	—	2006-03-08	
6-LA-103-04	7.11	3.56	—	—	7.17	—	—	—	2006-03-08	
6-LA-103-05	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-06	7.11	3.56	—	—	7.16	—	—	—	2006-03-08	
6-LA-103-07	7.11	3.56	—	—	7.18	—	—	—	2006-03-08	
6-LA-103-08	7.11	3.56	—	—	7.04	—	—	—	2006-03-08	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	—	—	8.52	—	—	—	2006-03-08	
4-RA-101-02	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-101-03	8.56	3.81	—	—	8.61	—	—	—	2006-03-08	
4-RA-101-04	8.56	3.81	—	—	8.63	—	—	—	2006-03-08	
4-RA-101-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-07	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-08	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	
4-RA-101-09	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-101-10	8.56	3.81	—	—	8.67	—	—	—	2006-03-08	
4-RA-101-11	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-02	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-102-03	8.56	3.81	—	—	8.56	—	—	—	2006-03-08	
4-RA-102-04	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-05	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-102-06	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-102-07	8.56	3.81	—	—	8.58	—	—	—	2006-03-08	
4-RA-102-08	8.56	3.81	—	—	8.49	—	—	—	2006-03-08	
4-RA-102-09	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	—	—	11.00	—	—	—	2006-03-08	
6-RA-103-02	10.97	4.57	—	—	11.09	—	—	—	2006-03-08	
6-RA-103-03	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-04	10.97	4.57	—	—	10.98	—	—	—	2006-03-08	
6-RA-103-05	10.97	4.57	—	—	10.95	—	—	—	2006-03-08	
6-RA-103-06	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-07	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-08	10.97	4.57	—	—	10.97	—	—	—	2006-03-08	
6-RA-103-09	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-10	10.97	4.57	—	—	11.02	—	—	—	2006-03-08	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-02	7.62	3.56	—	—	7.57	—	—	—	2001-03-08	
3-RV-101-03	7.62	3.56	—	—	7.52	—	—	—	2001-03-08	
3-RV-101-04	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-05	7.62	3.56	—	—	7.67	—	—	—	2001-03-08	
3-RV-101-06	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	—	—	5.53	—	—	—	2001-03-08	
2-RV-102-02	5.54	2.79	—	—	5.59	—	—	—	2001-03-08	
2-RV-102-03	5.54	2.79	—	—	5.48	—	—	—	2001-03-08	
2-RV-102-04	5.54	2.79	—	—	5.58	—	—	—	2001-03-08	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	—	—	5.48	—	—	—	2006-03-08	
3-RF-101-02	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-03	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-04	5.49	2.41	—	—	5.46	—	—	—	2006-03-08	
3-RF-101-05	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-06	5.49	2.41	—	—	5.52	—	—	—	2006-03-08	
3-RF-101-07	5.49	2.41	—	—	5.42	—	—	—	2006-03-08	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	—	—	3.92	—	—	—	2006-03-08	
2-RF-102-02	3.91	2.16	—	—	3.88	—	—	—	2006-03-08	
2-RF-102-03	3.91	2.16	—	—	3.93	—	—	—	2006-03-08	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2006-03-08	
2-RF-102-07	3.91	2.16	—	—	3.96	—	—	—	2006-03-08	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-104-02	6.02	3.18	—	—	6.03	—	—	—	2006-03-08	
4-LA-104-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-104-04	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-104-05	6.02	3.18	—	—	6.04	—	—	—	2006-03-08	
4-LA-104-06	6.02	3.18	—	—	6.10	—	—	—	2006-03-08	
4-LA-104-07	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-104-08	6.02	3.18	—	—	5.94	—	—	—	2006-03-08	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-104-02	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-104-03	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-104-04	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-104-06	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-07	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	—	—	3.93	—	—	—	2006-03-08	
2-ST-101-02	3.91	2.29	—	—	3.90	—	—	—	2006-03-08	
2-ST-101-03	3.91	2.29	—	—	3.94	—	—	—	2006-03-08	
2-ST-101-04	3.91	2.29	—	—	3.82	—	—	—	2006-03-08	
2-ST-101-05	3.91	2.29	—	—	3.91	—	—	—	2006-03-08	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	—	—	5.42	—	—	—	2006-03-08	
3-CW-101-02	5.49	2.79	—	—	5.41	—	—	—	2006-03-08	
3-CW-101-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-101-04	5.49	2.79	—	—	5.46	—	—	—	2006-03-08	
3-CW-101-05	5.49	2.79	—	—	5.48	—	—	—	2006-03-08	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	—	—	5.50	—	—	—	2006-03-08	
3-CW-102-02	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	
3-CW-102-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-102-04	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	1996-03-08	_____